HEC Montréal, Department of Finance

Granular Gravity: Equity-Bond Returns & Correlations

Ali

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Motivation

Distribution of firm size is heavy tailed

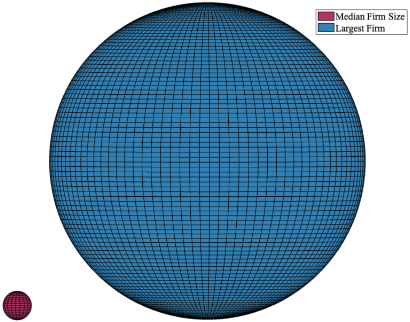

Figure 1: Largest versus Median Firm Size (2002-2019)

- Feb 2021: Biggest five tech firms, \$7.8Trillion; NYSE, \$26Trillion
- Largest firm 1400 bigger than median firm

Motivation

Granularity is well studied in Macroeconomics research

- Gabaix (2011, Econometrica) coined this phenomenon 'granularity'
- Granularity is not an anomaly, it is the true state of the world
- A growing body of literature documents the macroeconomic effects of granularity
- The core idea is that risk inherent in these entities is in-compressible
- Well, room for finance research

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Questions I pose in this paper

- **Is granularity informative within financial markets (equity and bond)?**

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- **Is granularity informative within financial markets (equity and bond)?**
- **Do co-movements across financial markets stem from granularity?**
- **Does theory support what we observe?**

Main Results

I'd like dessert served first!

- Granularity is a significant channel that impacts returns and correlations within and across financial markets
  - It is a priced factor in the cross section of equity and bond returns
  - As a priced factor, granularity impacts firms equity and bond return correlation (higher in high granularity months)
  - Theory supports these observations when the consumption basket is particularly dominated by grains

Literature

Granularity

- Gabaix (2011, Econometrica), di Giovanni and Levchenko (2012, Econometrica), Barrot and Sauvagnat (2016, QJE), Gabaix & Koijen (2019, WP), Carvalho and Grassi (2019, AER), Herscovic, Kelly, Lusting, and Van Nieuwerburgh (2020, JPE)

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- SML and equity prices versus granularity Abolghasemi, Bhamra, Dorion, Jeanneret (2019, WP)

Literature

Determinants of Equity-Bond Correlation (EBC)

Information versus risk: Kwan (1996, JFE), Acharya and Johnson (2007, JFE), Black and Crotty (2014, RFS), Bao and Pan (2013, RFS), Bethke, Gehde-Trapp, and Kempf (2017, RFE), Bektic (2018, JFI)

Literature

Cross-Sectional determinants of Bond Returns

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Cross-Sectional determinants of Bond Returns

- Traditionally, term and default spread (Fama and French, 1993, JFE)
- Bond-specific factors: Bai, Bali, and Wen (2019, JFE)
- Macroeconomic uncertainty (Bai, Subrahmanyam, and Wen, 2020b, JFQA) or uncertainty about credit ratings (Kim, Kim, and Park, 2018)
- Risk factors common with equity: (il)liquidity (Acharya, Amihud, and Bharath, 2013, JFE), momentum (Jostova, Nikolova, Philipov, and Stahel, 2013, RFS), reversal (Bali, Subrahmanyam, and Wen, 2020a, JFE)

Data

Equity & Bond Returns are from CRSP and Enhanced TRACE

- Daily bond transactions from Enhanced TRACE (2002-2019)
- Bond characteristics from Mergent FISD
- Filtering à la Bai, Bali, and Wen (2019) (Only bonds listed in US market, exclude convertible bonds, and bonds less than $5 or more than $1,000, with less than one year to maturity. More)
- Bond returns (1.4m month-return observations) are computed as

$$r_{b,t} = \frac{P_{b,t} + AI_{b,t} + C_{b,t}}{P_{t-1} + AI_{b,t-1}} - 1$$

$AI_{b,t}$ is accrued interest, and $C_{b,t}$ is the coupon.

- CRSP for equity data
- Matched 1,797 firms

Data

Granularity is measured similar to market concentration

- The value of the firm is $V_i = E_i + B_i$
- Define the value of the firm as a fraction of total market value: $w_i = \frac{V_i}{V_M}$
- The Herfindahl–Hirschman Index (HHI) is a widely used measure of market concentration:

$$HHI = \sum_{i=1}^{N} w_i^2 \tag{1}$$

- Gabaix and Koijen (2019) introduce the excess Herfindahl–Hirschman Index (exHHI),

$$exHHI = \sqrt{-\frac{1}{N} + \sum_{i=1}^{N} w_i^2} \tag{2}$$

Data

Alternative measures look quite similar

- Taking the value of top firms as fraction of total market value is highly correlated with the exHHI and HHI

$$Top_g = \frac{\sum_{j=1}^{g} V_j}{\sum_{i=1}^{N} V_i} \tag{3}$$

- g is the number of grains
- Telling grains from big firms is not straightforward, and g is an empirical estimate

Data

Time evolution of granularity (2002-2019), Corr(exHHI,Top_5)=0.978, Corr(ΔexHHI,ΔTop_5=0.922)

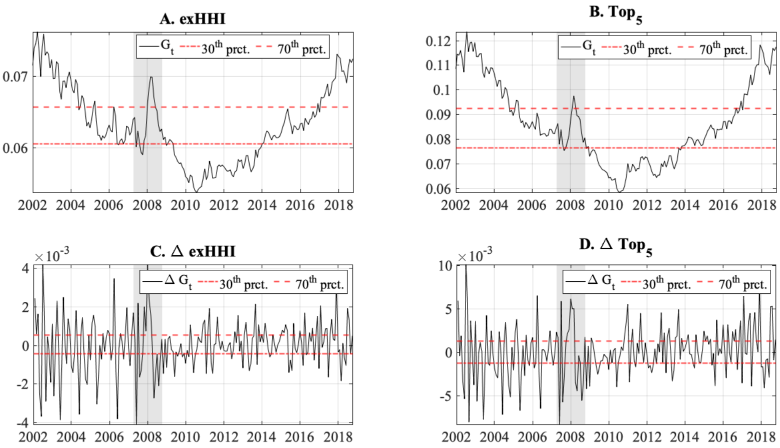

Figure 2: Granularity Measures

Motivating Empirics (Correlations)

Conditional distribution of EBC with respect to granularity

- Define conditional equity and bond returns correlation (EBC) as

$$\begin{aligned} \rho_h &= Corr(r_{b,t}, r_{e,t} | \Delta G_t > Q_{\Delta G,70}) \\ \rho_l &= Corr(r_{b,t}, r_{e,t} | \Delta G_t < Q_{\Delta G,30}) \end{aligned} \tag{4}$$

- It is informative to have the conditional distribution of average EBCs

Motivating Empirics (Correlations)

Construct a more balanced panel of returns for reliable estimations of EBC

I follow these steps to construct the distribution of the correlation coefficients introduced above

- Form 90 portfolios of random firms (20 firms in every portfolio)
- Compute the conditional correlations for each portfolio (resulting in 90 coefficients)
- Sample, 50,000 times, from the estimated coefficients and record the average of each sample
- Plot the histogram of the bootstrapped values

Motivating Empirics (Correlations)

Conditional distributions, $\Delta G = \Delta \text{exHHI}$

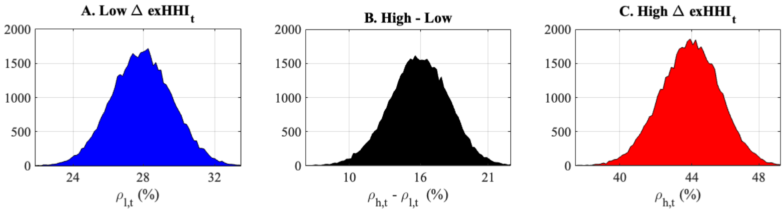

Figure 3: Conditional Distribution of correlations

Motivating Empirics (Correlations)

Conditional distributions, $\Delta G = \Delta \text{Top}_{20}$

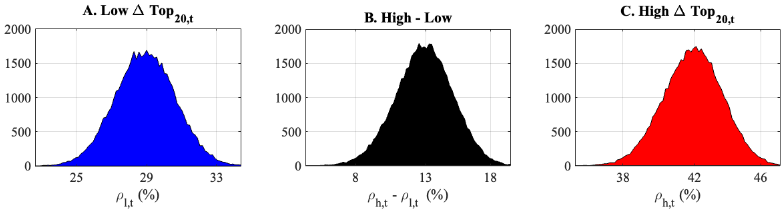

Figure 4: Conditional Distribution of correlations

Further Empirical Support

How rolling correlations relate to granularity

- As a more formal test, I compute rolling correlations, $w = 36$,

$$\hat{\rho}_{e,b,t} = \frac{\sum_{s=t-w-1}^{t-1} r_{e,s}, r_{b,s}}{\left(\sum_{s=t-w-1}^{t-1} r_{e,s}^2\right)\left(\sum_{s=t-w-1}^{t-1} r_{b,s}^2\right)} \tag{5}$$

- I estimate the following panel regressions,

$$\Delta\hat{\rho}_{e,b,t} = b_0 + \beta_g \Delta G_t + \sum_{c=1}^{C} \beta_c Control_{c,t} + \eta_t \tag{6}$$

Further Empirical Support

Regress $\Delta\rho$ on ΔTop_{20}, 2002-2019 ⏩ Grain Returns

Var.	ΔG_t^{Top}	R_{Top-R_f}	Controls
$\Delta\hat{\rho}$	0.035*** (10.30)	0.002 (0.70)	N
$\Delta\hat{\rho}$	0.055*** (12.85)	0.004 (0.37)	Y

Table 1: Fixed Effects - Changes in Correlation versus Granularity, $\Delta \text{G} = \Delta \text{Top}_{20}$

Theory

Explain how granularity relates to returns and correlations

This theoretical framework should explain why granularity is

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O3. Positively related to return on grains

O4. Impacts individual equity and bond returns in the same direction

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Assumptions to build a granular economy

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A4. A representative agent with log utility prices assets

A5. The cashflow of grains is less volatile than the cashflow of non-grains, $\sigma_g < \sigma_i$

Theory

Consumption growth is key to pricing assets

The representative agent with log utility consumes the aggregate cashflow,

$$C_t = \sum_{i=1}^{N} X_i, \tag{9}$$

hence consumption growth is

$$\frac{dC_t}{C_t} = \sum_{i=1}^{N} w_i \frac{dX_i}{X_i}, \tag{10}$$

where w_i is the share of firm i in the consumption basket.

Theory

Consumption growth is driven by a market-wide as well as firm related components

By A3 $\left(dB_i = \rho_i dZ_M + \sqrt{1-\rho_i^2} dZ_i \right)$,

$$\frac{dX_i}{X_i} = \mu_i dt + \sigma_i \left(\rho_i dZ_M + \sqrt{1-\rho_i^2} dZ_i\right), \tag{11}$$

then consumption growth

$$\begin{aligned} \frac{dC_t}{C_t} &= \sum_{i=1}^{N} w_i \frac{dX_i}{X_i} \\ &= \sum_{i=1}^{N} w_i \mu_i dt + \underbrace{\sum_{i=1}^{N} w_i \sigma_i \rho_i dZ_M}_{\text{Market-wide component}} + \underbrace{\sum_{i=1}^{N} w_i \sigma_i \sqrt{1-\rho_i^2} dZ_i}_{\text{firm specific component}} \end{aligned} \tag{12}$$

- The Stochastic Discount Factor is governed by

$$\frac{d\pi_t}{\pi_t} = -r_f dt + \sigma_M dZ_M + \sum_{i=1}^{N} w_i \sqrt{(1-\rho_i^2)}\sigma_i dZ_i \tag{13}$$

- where the risk free interest rate, r_f, is determined endogenously
- The cashflow risk for firm i is

$$\mathbb{E}\left[\frac{d\pi_t}{\pi_t}, \frac{dX_i}{X_i}\right] = \rho_i \sigma_i \sigma_M + w_i (1-\rho_i^2)\sigma_i^2 \tag{14}$$

Theory

Dynamics of equity and bond returns

- But we are interested in equity and bond risk premia
- By Ito's lemma, equity and bond returns are governed by

$$\frac{dE_t}{E_t} = \mu_e dt + \rho_i \sigma_e dZ_M + \sqrt{1-\rho_i^2}\sigma_e dZ_i, \tag{15}$$

and

$$\frac{dB_t}{B_t} = \mu_b dt + \rho_i \sigma_b dZ_M + \sqrt{1-\rho_i^2}\sigma_b dZ_i, \tag{16}$$

Theory

Debt and Equity risk premia follow similarly

- And corresponding risk premia,

$$\mathbb{E}\Big[\frac{d\pi_t}{\pi_t}, \frac{dE_t}{E_t}\Big] = \rho_i \sigma_e \sigma_M + w_i (1 - \rho_i^2) \sigma_e \sigma_i, \tag{17}$$

and

$$\mathbb{E}\Big[\frac{d\pi_t}{\pi_t}, \frac{dB_t}{B_t}\Big] = \rho_i \sigma_b \sigma_M + w_i (1 - \rho_i^2) \sigma_b \sigma_i, \tag{18}$$

- Solve for σ_e and σ_b when firms issue an optimal mix of debt and equity (Leland, 1994)
- For an economy with three firms, one grain, and two non-grains
- The only variable is the $w_g = \frac{X_g}{C}$

Theory

Firms riskiness when ρ_i is constant for all firms

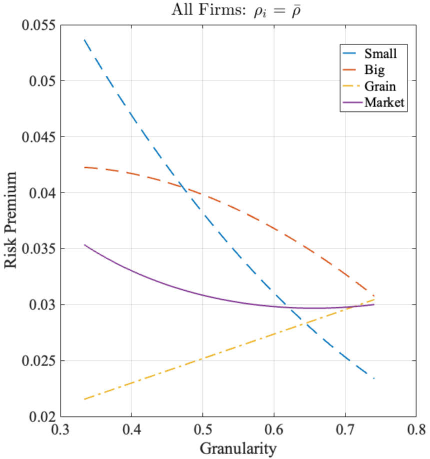


Figure 5: Model Implied Cashflow Risk Premia

Theory

Firms riskiness in different scenarios for how ρ_i behaves

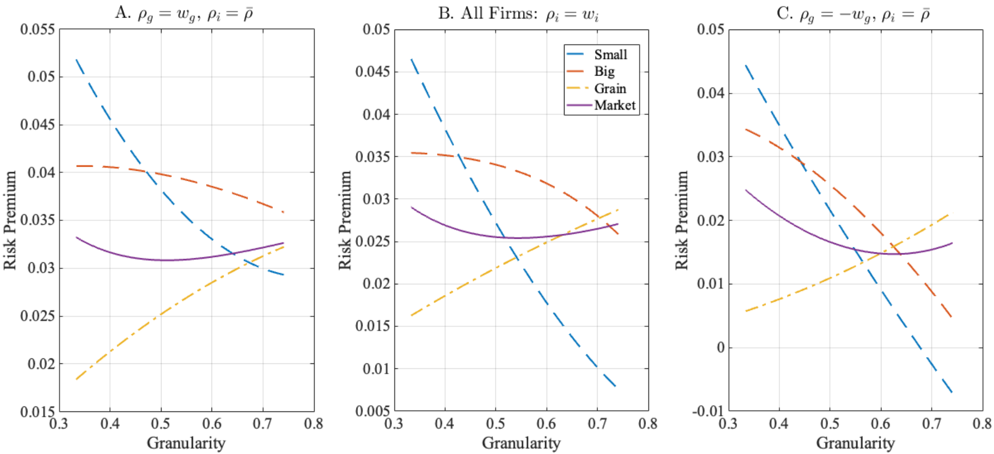


Figure 6: Model Implied Cashflow Risk Premia

Security Risk Premia ⏩ Other Scenarios

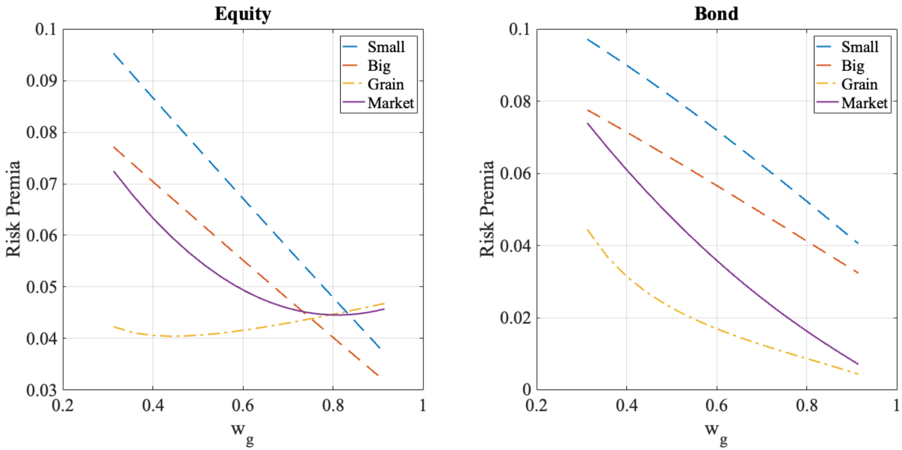


Figure 7: Model Implied Equity and Bond Risk Premia

Empirical Support

FM Regressions: Is granularity priced in CS of firm securities?

- The model implies that granularity is negatively related to return on non-grains
- I examine the relevance of granularity in the CS of equity and bond returns by estimating a two-pass Fama-Macbeth regression
- I control for all common risk factors in the cross section of equity and bond returns (Bond market return, default and term spreads, financial uncertainty (Jurado et al., 2015), liquidity, downside, and credit risk factors of Bai et al. (2019); For the equity market, Fama and French five factors, momentum, and financial uncertainty.

Empirical Support

FM Regressions, ΔTop$_{20}$, Monthly Equity and Bond Returns (2002-2019)

Var.	$\gamma_{\Delta G_t^{Top}}$	$\gamma_{R_{Top}-R_f}$	Controls
Bond	-0.114**	0.121***	N
	(-2.09)	(3.23)	
	-0.171***	0.165***	Y
	(-3.08)	(3.07)	
Equity	-0.082**	0.065	N
	(-2.09)	(1.22)	
	-0.102***	0.044	Y
	(-3.19)	(0.98)	

Table 2: Fama-Macbeth CS Regressions, Individual Securities, $\Delta G = \Delta \text{Top}_{20}$

Empirical Support

Fixed Effect Regressions, ΔTop_{20}, 2002-2019

Var.	$\beta_{\Delta G_t^{Top}}$	$\beta_{R_{Top}-R_f}$	Controls
Bond, Firm Level	-0.013**	0.024***	N
	(-4.61)	(7.77)	
	-0.019***	0.024***	Y
	(-6.09)	(6.81)	
Bond, Matched Firms	-0.031***	0.045***	N
	(-3.58)	(6.53)	
	-0.066***	0.067***	Y
	(-7.72)	(6.87)	

Table 3: Fixed Effect Regressions, $\Delta G = \Delta\text{Top}_{20}$

Conclusion

- Risk inherent in large firms grows systematic
- Granularity is priced in the cross section of equity and bond returns
- The Equity-Bond correlation is therefore higher when financial markets are more granular

Thank You!

Appendix

Cleaning Criteria

Filters à la Bai, Bali, and Wen (2019)

I first apply the procedure suggested in Nielsen (2014) and then apply the following filters on remaining transactions:

- Remove bonds that are not listed or traded in the US public market, which include bonds issued through private placement, bonds issued under the 144A rule, bonds that do not trade in US dollars, and bond issuers not in the jurisdiction of the United States.
- Remove bonds that are structured notes, mortgage backed or asset backed, agency backed, or equity linked.
- Remove convertible bonds since this option feature dis- torts the return calculation and makes it impossible to compare the returns of convertible and nonconvertible bonds
- Remove bonds that trade under \$5 or above \$1,000.

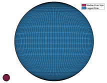
Cleaning Criteria

Filters à la Bai, Bali, and Wen (2019)

- Remove bonds that have a floating coupon rate, which means the sample comprises only bonds with a fixed or zero coupon.
- Remove bonds that have less than one year to maturity.
- Eliminate bond transactions that are labeled as when-issued, locked-in, or have special sales conditions, and that have more than a two- day settlement.
- Remove transaction records that are canceled and ad- just records that are subsequently corrected or re- versed.
- Remove transaction records that have trading volume less than $10,000.

Back

Granularity is different from return on Top firms



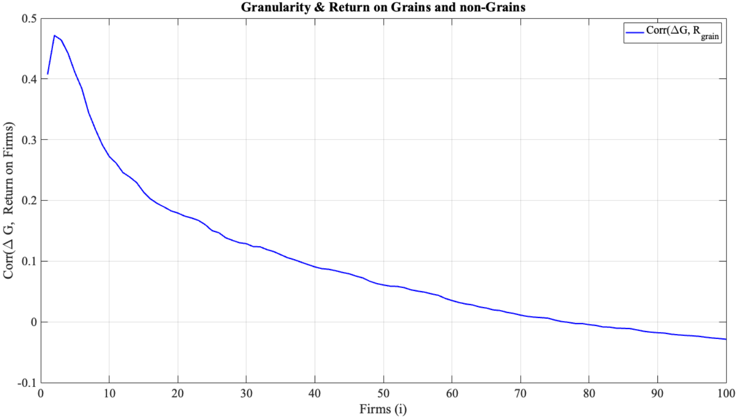


Figure 8: Granularity versus Returns

Robustness check

Granular shocks are negatively related to return on non-grains

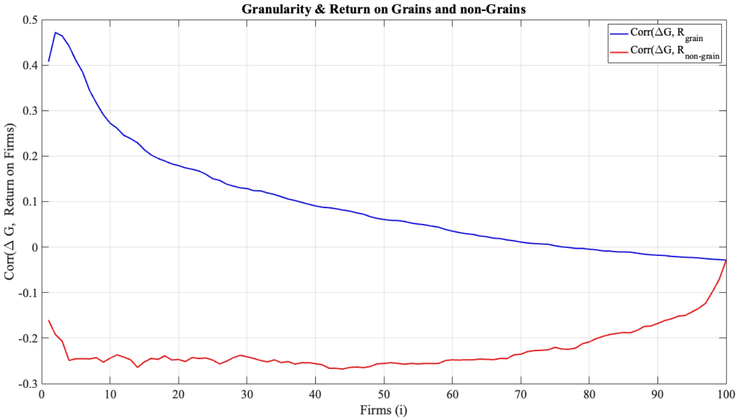


Figure 9: Granularity versus Returns

Robustness check

Granular shocks are different from market return

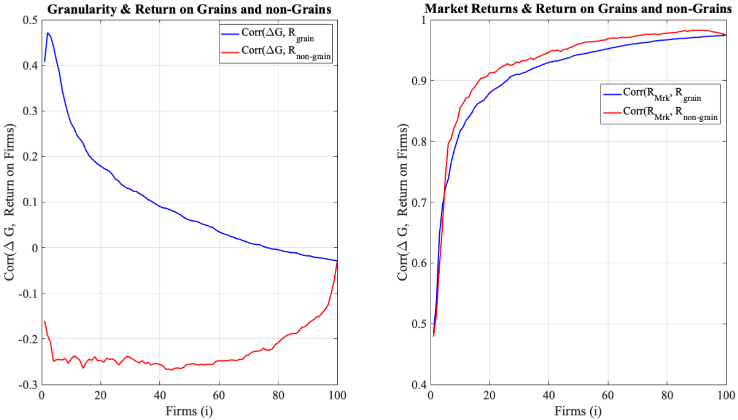
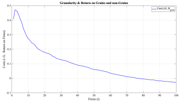


Figure 10: Granularity versus Returns

Various Measures of Granularity



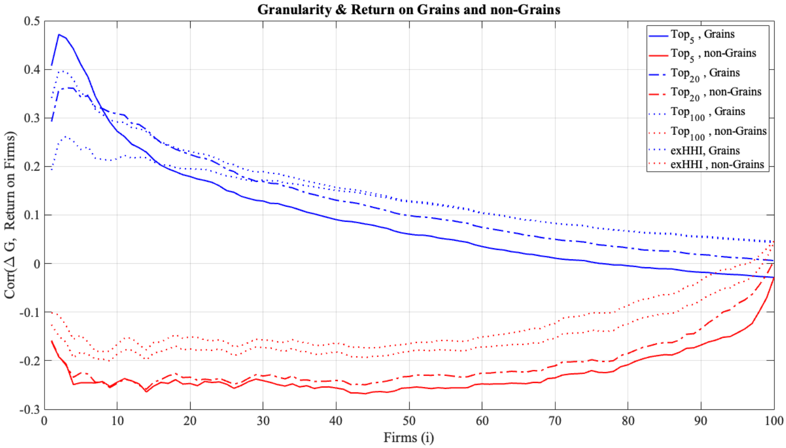


Figure 11: Granularity versus Returns Back

Robustness check

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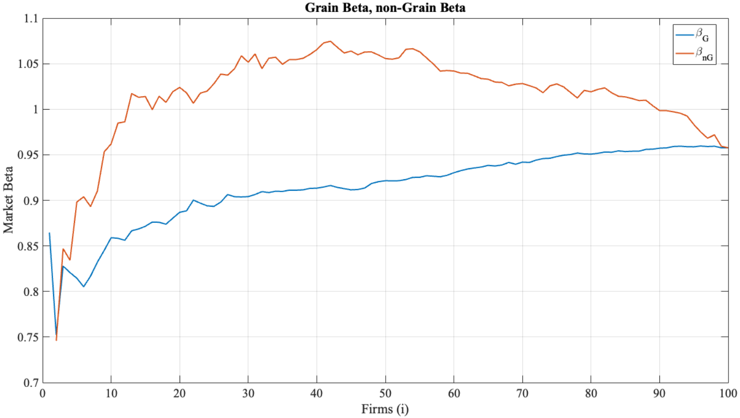

Figure 12: Market Betas for Grains ans non-Grains

Complementary: Theory

Security Risk Premia, $\rho_i = w_i$

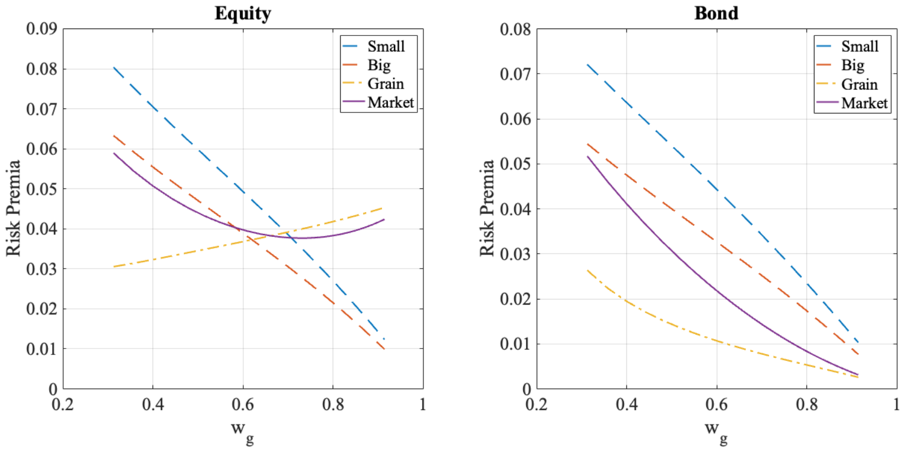

Figure 13: Model Implied Equity and Bond Risk Premia

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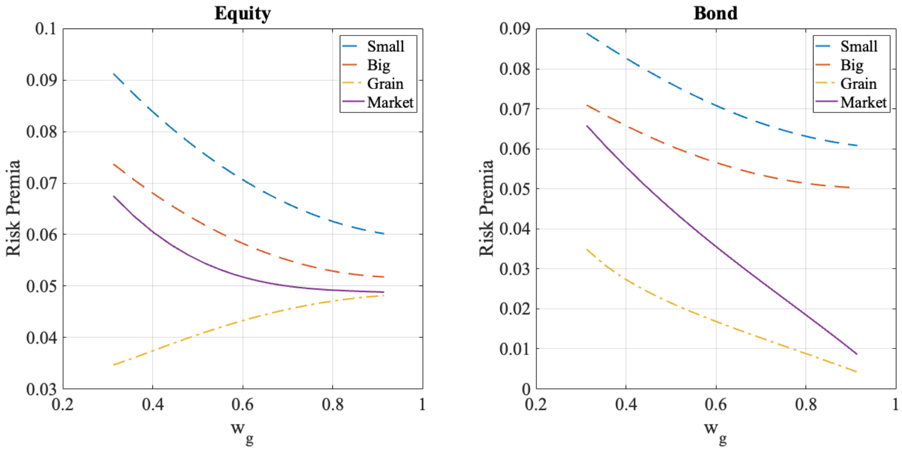

Figure 14: Model Implied Equity and Bond Risk Premia Back